

Statistically Valid Inferences from Privacy Protected Data¹

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Google, 3/20/2020

¹Joint work with Georgina Evans, Margaret Schwenzfeier, Abhradeep Thakurta.

²[GaryKing.org/dp](https://garyking.org/dp)

Solving Political Problems Technologically

Differential Privacy & Inferential Validity

A General Purpose, Statistically Valid DP Algorithm

The Algorithm in Practice

Convincing Facebook to Make Data Available

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Solving a Political Problem Technologically (via “constitutional design”)

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- [New Problem](#): **Sharing data without it leaving Facebook**

Data Sharing Regime $\leadsto$ Data Access Regime

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Theories of Inference: Statistics vs. CS

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Population

⋮

Michele

Daniel

Georgie

Gary

Meg

Abhradeep

Scott

Larsy

Keith

Elise

Mean
income:

\$48

Quantity
of Interest

Theories of Inference: Statistics vs. CS

Population	Sample
:	X
Michele	✓
Daniel	✓
Georgie	✓
Gary	✓
Meg	✓
Abhradeep	✓
Scott	✓
Larsy	✓
Keith	✓
Elise	✓

Mean
income:

\$48

Quantity
of Interest

Theories of Inference: Statistics vs. CS

Population	Sample	\$
:	X	
Michele	✓	76
Daniel	✓	122
Georgie	✓	145
Gary	✓	96
Meg	✓	86
Abhradeep	✓	127
Scott	✓	72
Larsy	✓	132
Keith	✓	95
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Mean
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Classical
Inference

\$108

Quantity
of Interest

Usually
no direct
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Mean
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\$108

Quantity
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Theories of Inference: Statistics vs. CS

	Population	Sample	\$	+Privacy
	:	X		
	Michele	✓	76	Noise & Censoring
	Daniel	✓	122	
	Georgie	✓	145	
	Gary	✓	96	
	Meg	✓	86	
	Abhradeep	✓	127	
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Mean income:	\$48	Classical Inference		\$108
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Mean
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Query-
Response

\$111

Quantity
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Classical Inference

Query-Response

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Differential Privacy and its Inferential Challenges

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- Estimators

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 - Classical Statistics: Apply statistic s to dataset D , $s(D)$

Differential Privacy and its Inferential Challenges

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- **Statistical properties:** **usually biased**, **no uncertainty estimates**

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 - **Can address with:** careful software design & education

Solving Political Problems Technologically

Differential Privacy & Inferential Validity

A General Purpose, Statistically Valid DP Algorithm

The Algorithm in Practice

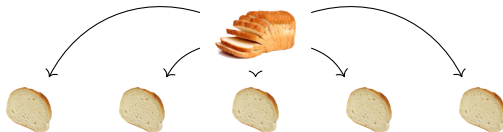
A Differentially Private Estimator

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Private data

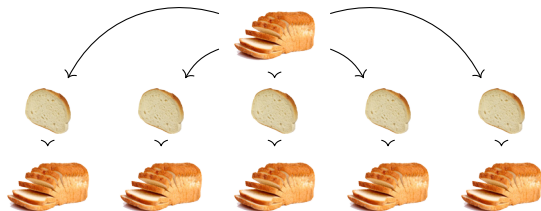
A Differentially Private Estimator



Private data

Partition

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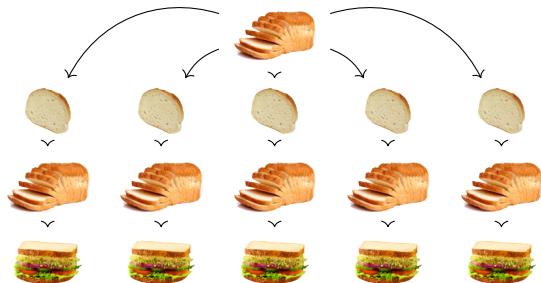


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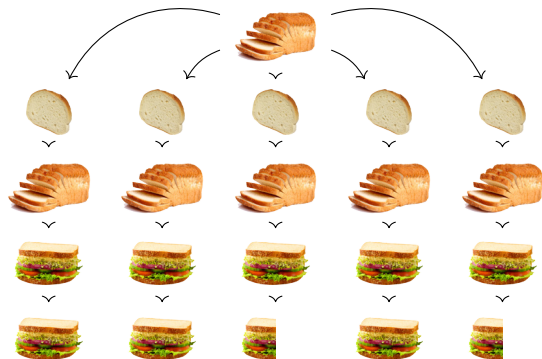
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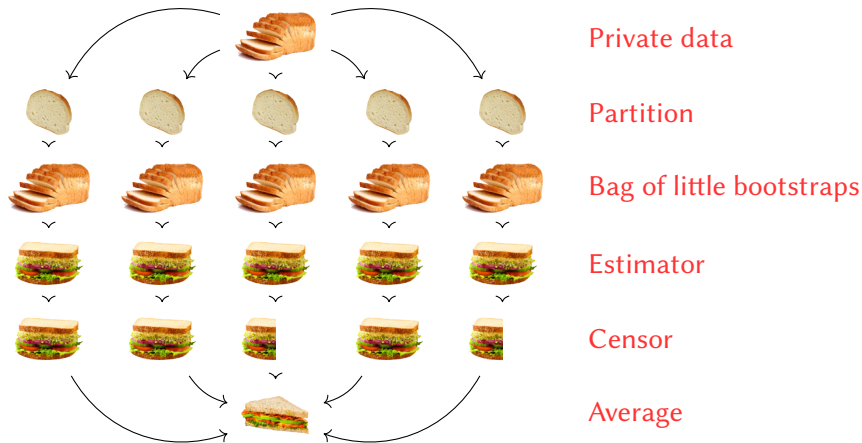
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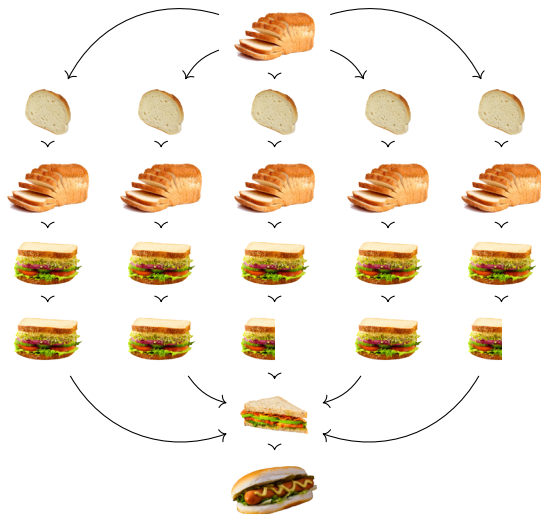
Estimator

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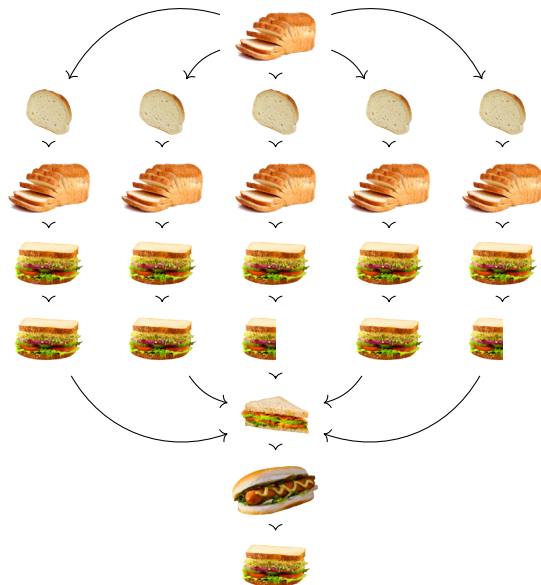
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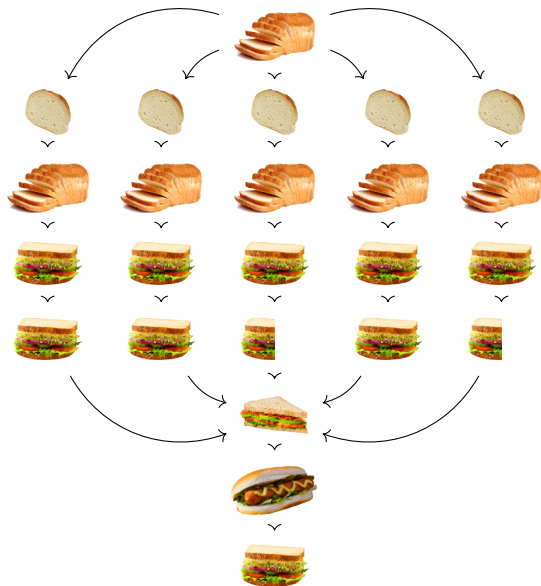
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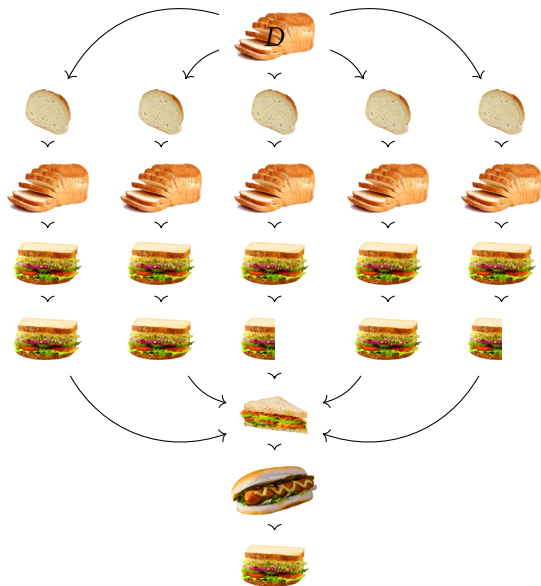
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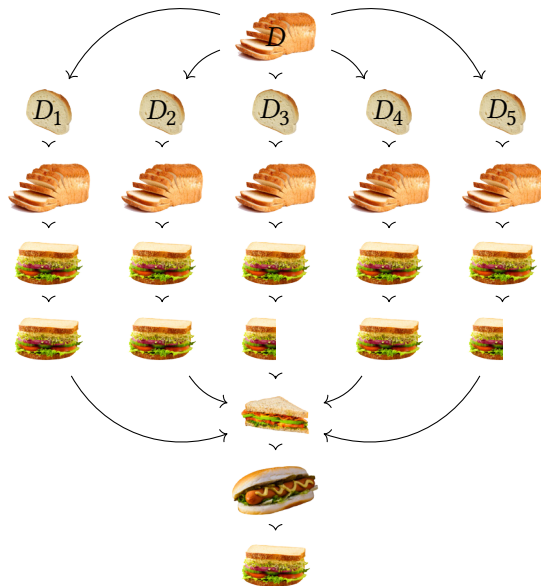
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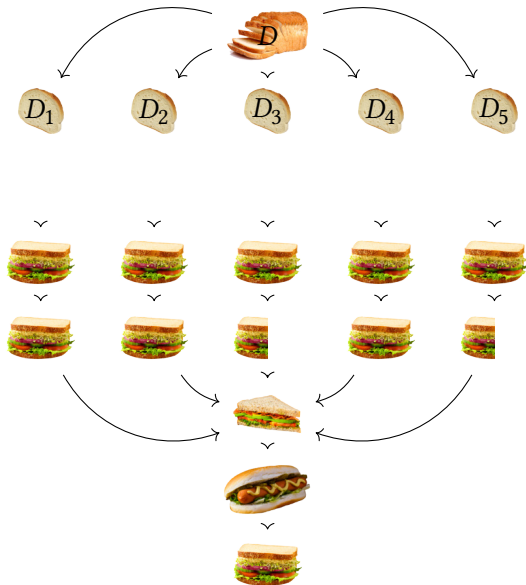
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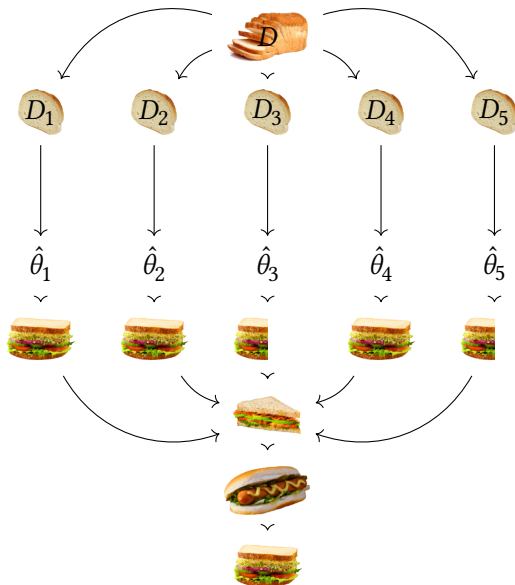
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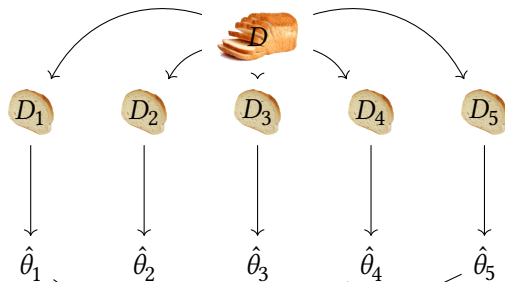
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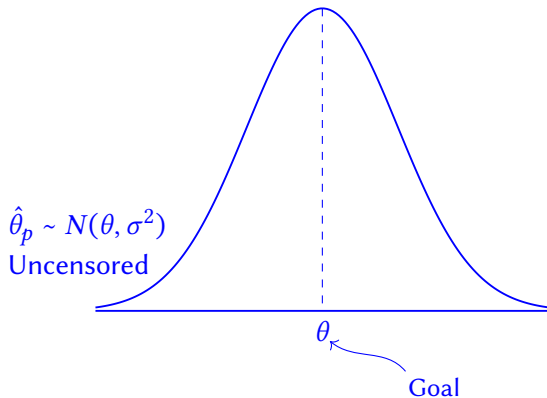
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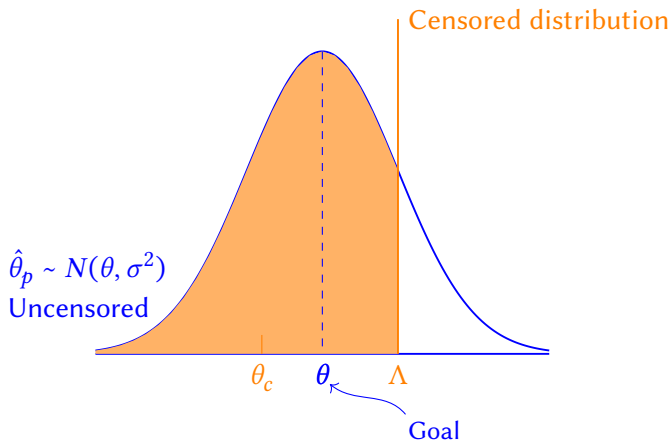
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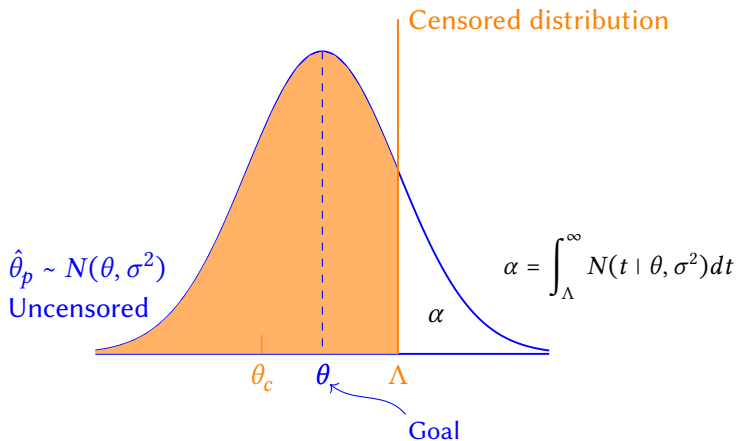
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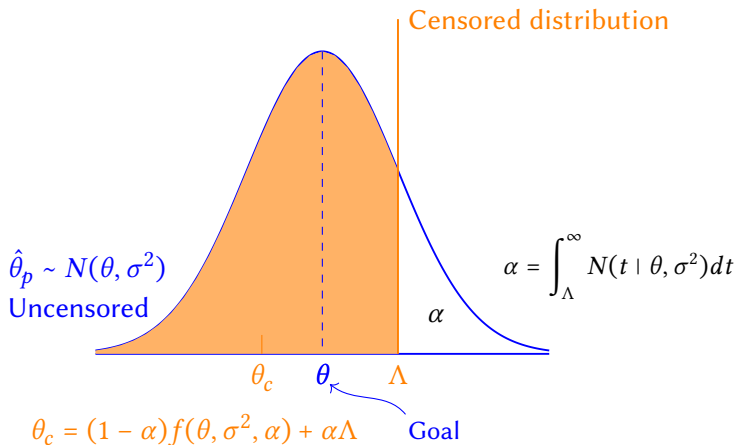
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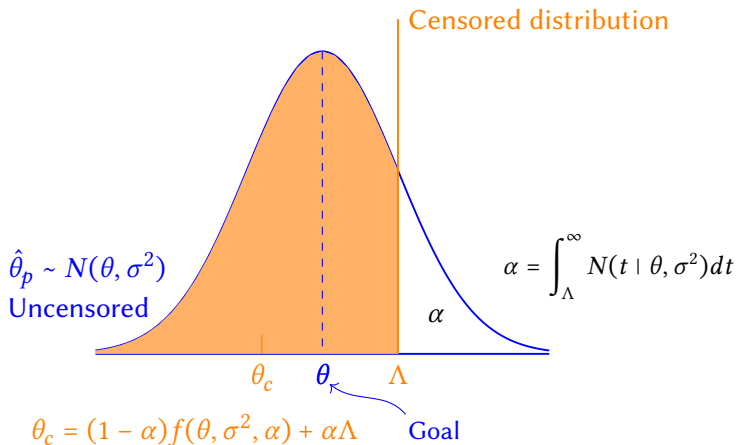
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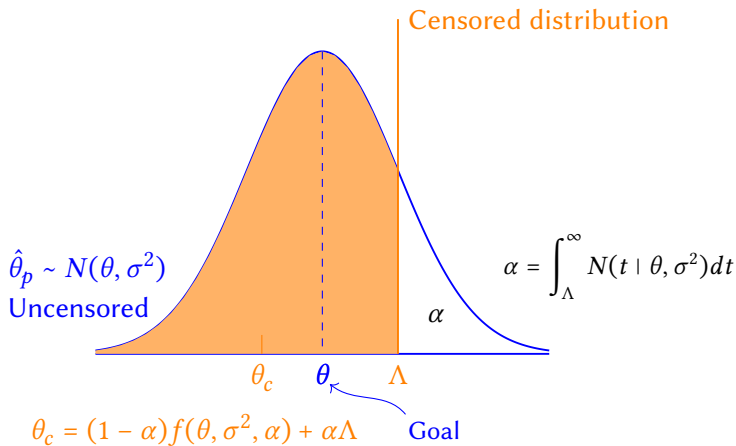


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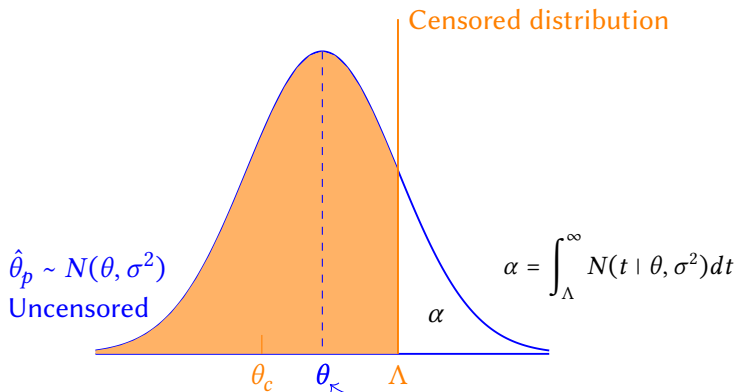
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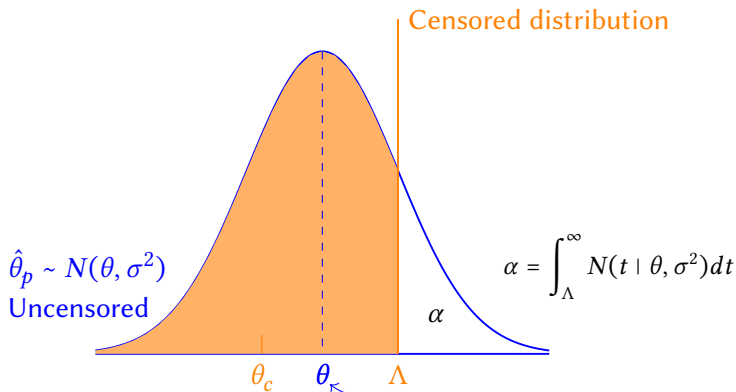
Goal

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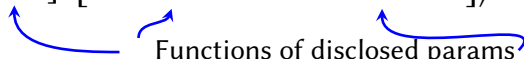
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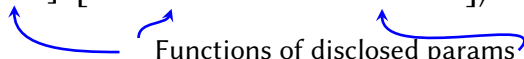
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- **Bias correction**: reduces bias *and* variance:

$$E(\tilde{\theta}^{\text{dp}}) \approx \theta, \quad V(\tilde{\theta}^{\text{dp}}) \lesssim V(\hat{\theta}^{\text{dp}})$$

Solving Political Problems Technologically

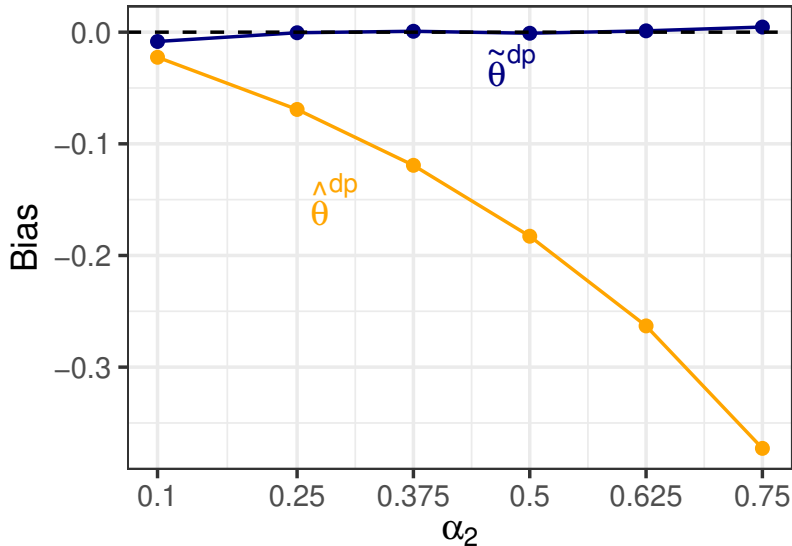
Differential Privacy & Inferential Validity

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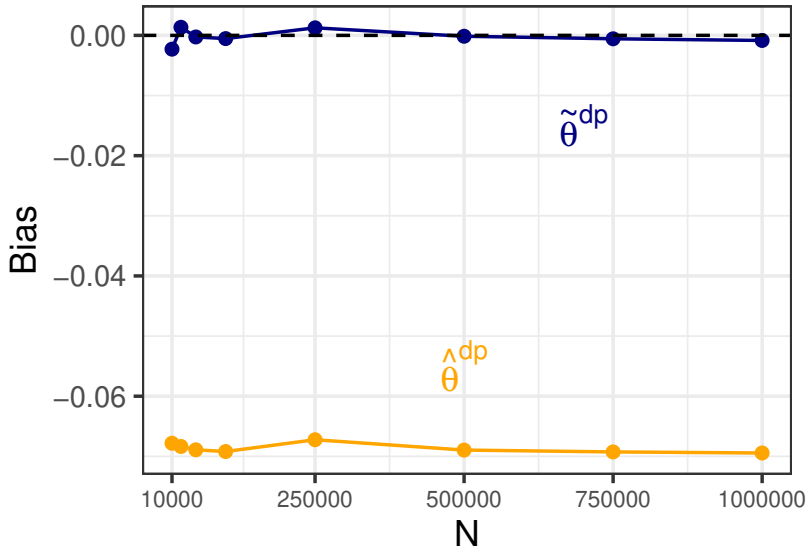
The Algorithm in Practice

Simulations: Finite Sample Evaluation

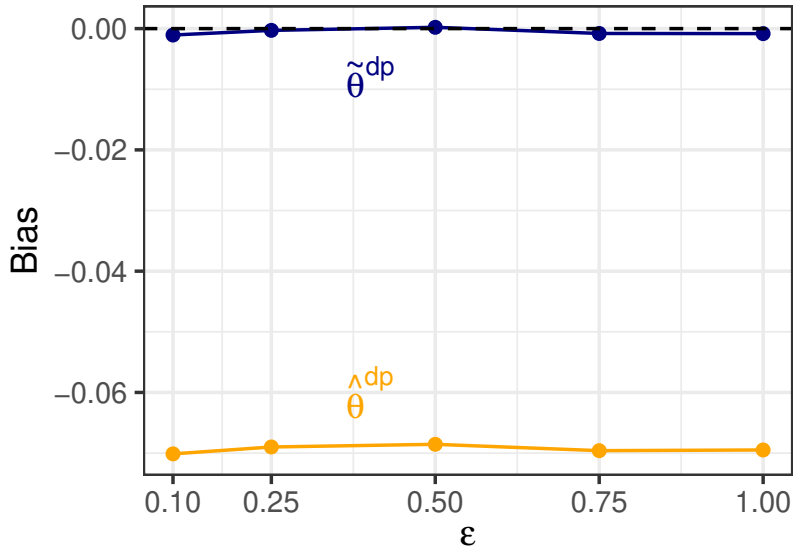
Simulations: Finite Sample Evaluation



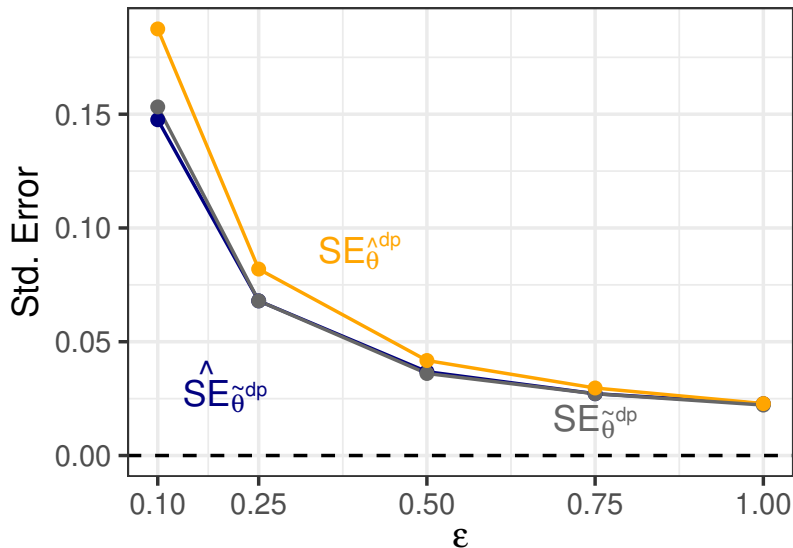
Simulations: Finite Sample Evaluation



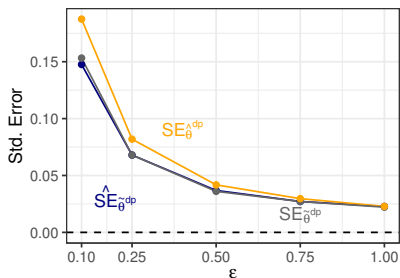
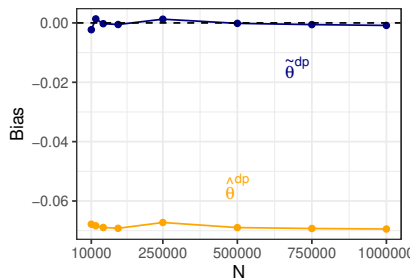
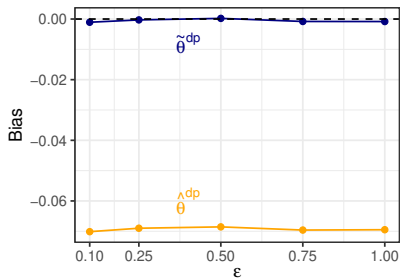
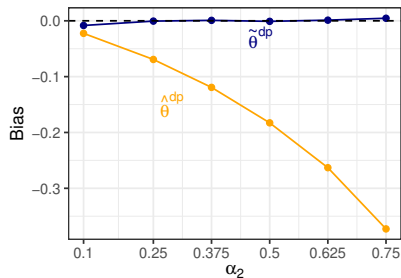
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